

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

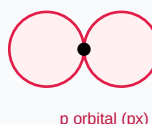
CHAPTER 13: ORGANIC NITROGEN COMPOUNDS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Organic Nitrogen Compounds

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

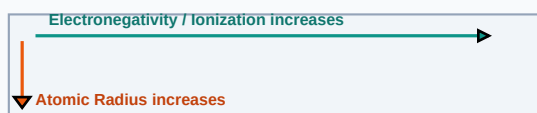
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Gabriel Phthalimide Synthesis	A chemical method used to prepare pure primary aliphatic amines without secondary/tertiary contamination.
Hoffmann Bromamide Degradation	The conversion of an amide to a primary amine with one less carbon atom, using bromine and alkali.
Diazotisation	The reaction of primary aromatic amines with nitrous acid at 0-5°C to form diazonium salts.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

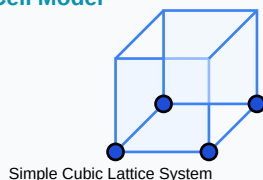
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Hoffmann Degradation: $\text{R-CONH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{R-NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$
- Amine Basicity: Secondary > Primary > Tertiary (in aqueous methylamines)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

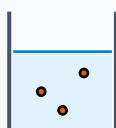
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Convert Aniline to Benzenediazonium Chloride.

Step-by-step Solution:

React Aniline ($\text{C}_6\text{H}_5\text{NH}_2$) with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) at ice cold temperature (273-278K). Diazotisation yields $\text{C}_6\text{H}_5\text{N}_2^+ \text{Cl}^-$.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

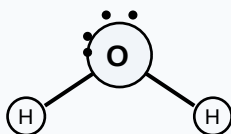
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain why aliphatic amines are stronger bases than aromatic amines like aniline.

Analytical Solution Strategy:

Aliphatic amines have +I alkyl groups which release electrons, increasing electron density on nitrogen. In aniline, the lone pair on nitrogen is delocalized into the benzene ring through resonance, making it less available for protonation.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

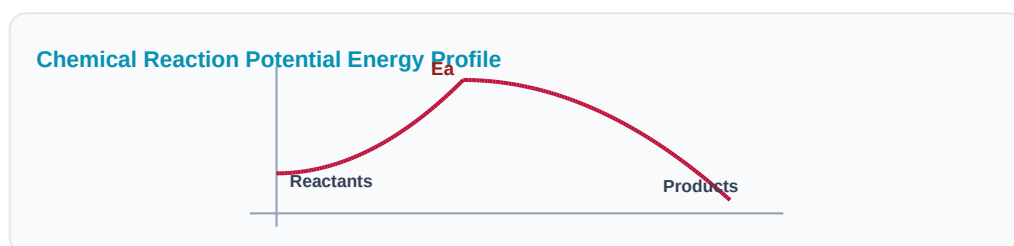
Q1 (1-Mark): Describe the preparation of primary amines by Gabriel Phthalimide Synthesis.

Q2 (2-Mark): Write chemical equations for: (1) Carbylamine reaction, (2) Mustard oil reaction.

Q3 (2-Mark): Explain the coupling reactions of benzenediazonium chloride with phenol.

Q4 (5-Mark): How do primary, secondary, and tertiary amines react with nitrous acid?

Q5 (5-Mark): Write a note on Schotten-Baumann reaction.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Dye Industry

Diazonium coupling reactions synthesize vibrant azo dyes for clothing textiles.

Application Case: Drug Design

Amine salts form the soluble base of local anesthetics like novocaine.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

